

Initial Design Document

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Segment 2 — Data Platform & Pipeline Engineering · Problem B1 · 24 June 2026

Project

CartCo Unified Commerce Lakehouse — a bronze → silver → gold medallion data pipeline for a multi-channel retailer whose revenue numbers differ across every team because data lives in five disconnected systems.

Problem Being Solved

CartCo sells across its own website, Amazon, Flipkart, Myntra, and 80 physical stores. Inventory, orders, and revenue data sit in Shopify, Seller APIs, MySQL POS, Kafka, and daily CSV drops from distributors. There is no single source of truth. This project builds the foundational data platform that unifies all of it into one trusted, queryable layer.

What I Will Build

Layer	What it does
Bronze	Raw ingestion from 3 sources, partitioned by date, schema-on-read
Silver	Cleaned, deduplicated, schema-enforced canonical tables
Gold	Business marts — daily revenue, channel performance, inventory
Orchestration	Airflow DAGs with retries, idempotency, and alerts
Data Quality	Great Expectations checks on critical silver and gold tables
Lineage	OpenLineage + Marquez for full pipeline traceability
Infra	Docker Compose — reproducible local stack, cloud deploy in Week 4

Tech Stack

Component	Choice	Reason
Object storage	MinIO	S3-compatible, runs locally via Docker, zero cost
Table format	Apache Iceberg	ACID, time-travel, schema evolution — modern lakehouse standard
Compute	PySpark	Industry standard for batch transforms, Python-friendly
Orchestration	Apache Airflow	DAG-based scheduling, familiar in my stack
Data quality	Great Expectations	Fail/warn checks on pipeline outputs
Lineage	OpenLineage + Marquez	Column-level traceability and pipeline visibility
IaC (stretch)	Terraform	Reproducible cloud provisioning if time permits

Scope

In scope: 3 synthetic source connectors · full medallion pipeline · Airflow DAGs · DQ suite · lineage · Docker Compose deployment.

Out of scope: Real-time sub-second streaming · multi-region HA · more than 3 sources in the initial build.

Bonus if ahead: Kafka streaming source · data contracts · Feast feature store.

Week-by-Week Plan

Week	Goal
Week 1 (done by 3 Jul)	Stack running · synthetic data · bronze layer ingested and queryable
Week 2 (8–11 Jul)	Silver transforms · gold marts · first end-to-end pipeline run · ADR-001
Week 3 (13–18 Jul)	Tests + CI · DQ suite · lineage · 3 ADRs total · Milestone 1 on 19 Jul
Week 4 (20–25 Jul)	Deploy · Loom · thinking artifact · resume bullets · Milestone 2 + Final

Key Risks and Mitigations

Risk	Mitigation
Tooling complexity — Spark + Iceberg + Airflow together	Integrate one component at a time; Docker Compose first
Scope creep toward streaming before batch is solid	Finish clean bronze → gold pipeline before adding extras

The goal of this project is not to build a perfect enterprise platform in five weeks. It is to build a disciplined, well-scoped, technically honest system that works end to end, is clearly documented, and reflects the way a real data engineering project would be approached under time constraints. That is the standard this design is aiming for.